

ADVANCED SOFTWARE ENGINEERING

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SCRUM

OUTLINE

- ▣ Definition
- ▣ **Roles :**
 - ▣ Product Owner,
 - ▣ ScrumMaster,
 - ▣ Team.
- ▣ Sprint
- ▣ How Scrum Works?
- ▣ **Artifacts**
 - Product Backlog,
 - Sprint Goals
 - Sprint Backlog,
 - and Burndown Chart
- ▣ **The Process:**
 - Sprint Planning,
 - Sprint Review,
 - Sprint presentation
 - Daily Scrum Meeting

Definition

- ▣ Scrum is an agile, lightweight process that can be used to manage and control software and product development using iterative, incremental practices

Definition

- ▣ Scrum is a process that allows us to focus on delivering the highest business value in the shortest time.
- ▣ It allows us to rapidly and repeatedly check actual working software (every two weeks to one month).

Definition

The business sets the priorities. Our teams self-manage to determine the best way to deliver the highest priority features.

Every two weeks to a month anyone can see real working software and decide to release it as is or continue to enhance for another iteration.

Characteristics of scum

- ▣ Self-organizing teams
- ▣ Product progresses in a series of month-long “sprints”
- ▣ Requirements are captured as items in a list of “product backlog”
- ▣ No specific engineering practices prescribed
- ▣ One of the “agile processes”

Components of Scrum

- ▣ **Roles** : Product Owner, ScrumMaster, Team.
- ▣ **Artifacts** : Product Backlog, Sprint Backlog, and Burndown Chart.
- ▣ **The Process**: Sprint Planning, Sprint Review, Sprint presentation and Daily Scrum Meeting

Roles

▣ **Roles :**

- 1)Product Owner,
- 2) ScrumMaster,
- 3) Team.

Product Owner

- ▣ Typically a Product Manager, Marketing, Internal Customer, etc.
- ▣ Define the features of the product
- ▣ Decide on release date and content
- ▣ Be responsible for the profitability of the product (ROI)
- ▣ Prioritize features according to market value
- ▣ Adjust features and priority every iteration, as needed
- ▣ Accept or reject work results.

The Scrum Master

- ▣ Represents management to the project
- ▣ Responsible for performing Scrum values and practices
- ▣ Removes obstacles
- ▣ Ensure that the team is fully functional and productive
- ▣ Enable close cooperation across all roles and functions
- ▣ Protect the team from external interferences

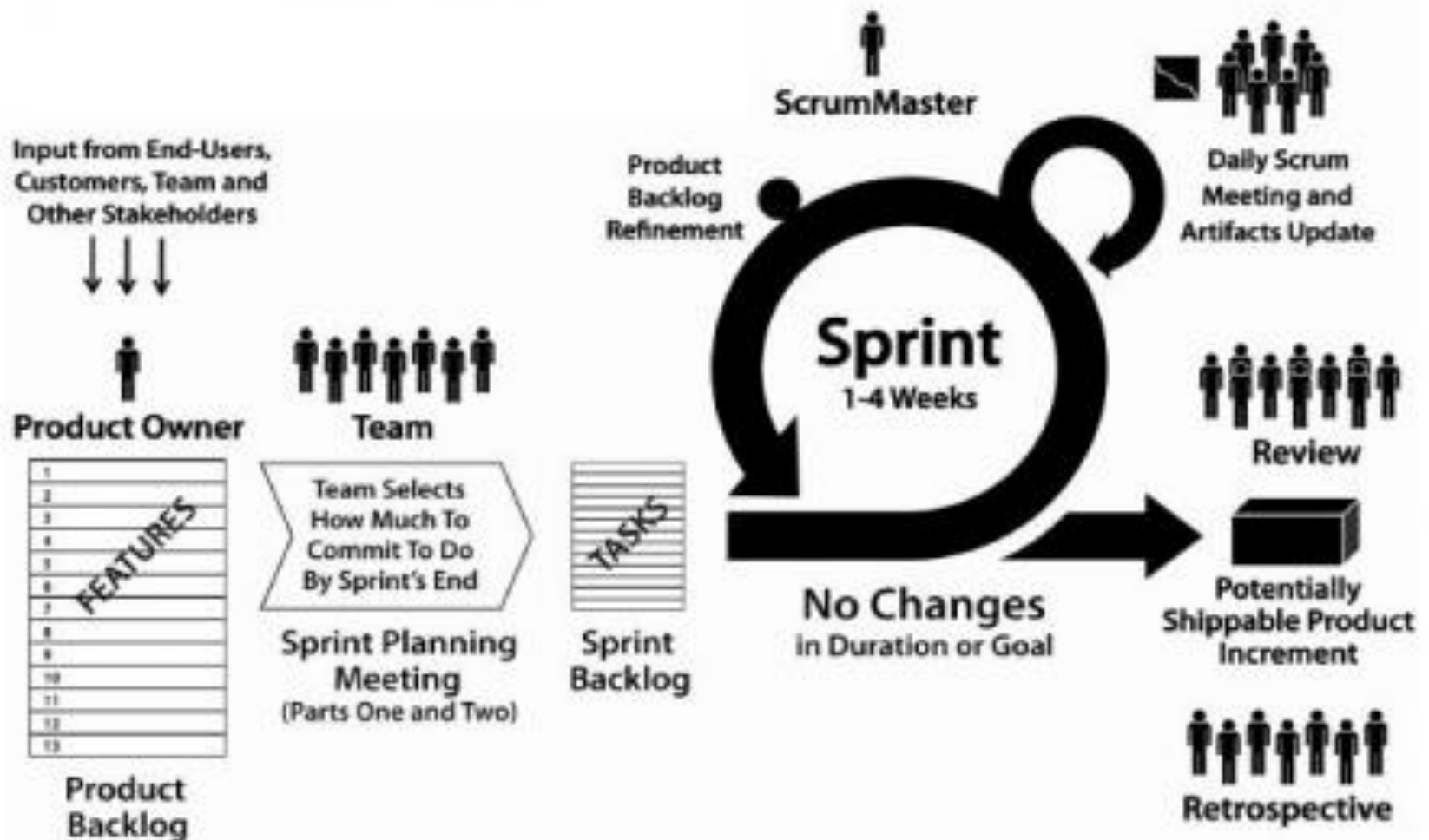
Scrum Team

- ▣ Typically 5-10 people
- ▣ Cross-functional
 - QA, Programmers, UI Designers, etc.
- ▣ Members should be full-time
 - May be exceptions (e.g., System Admin, etc.)
- ▣ Teams are self-organizing
 - What to do if a team self-organizes someone off the team??
 - Ideally, no titles but rarely a possibility
- ▣ Membership can change only between sprints

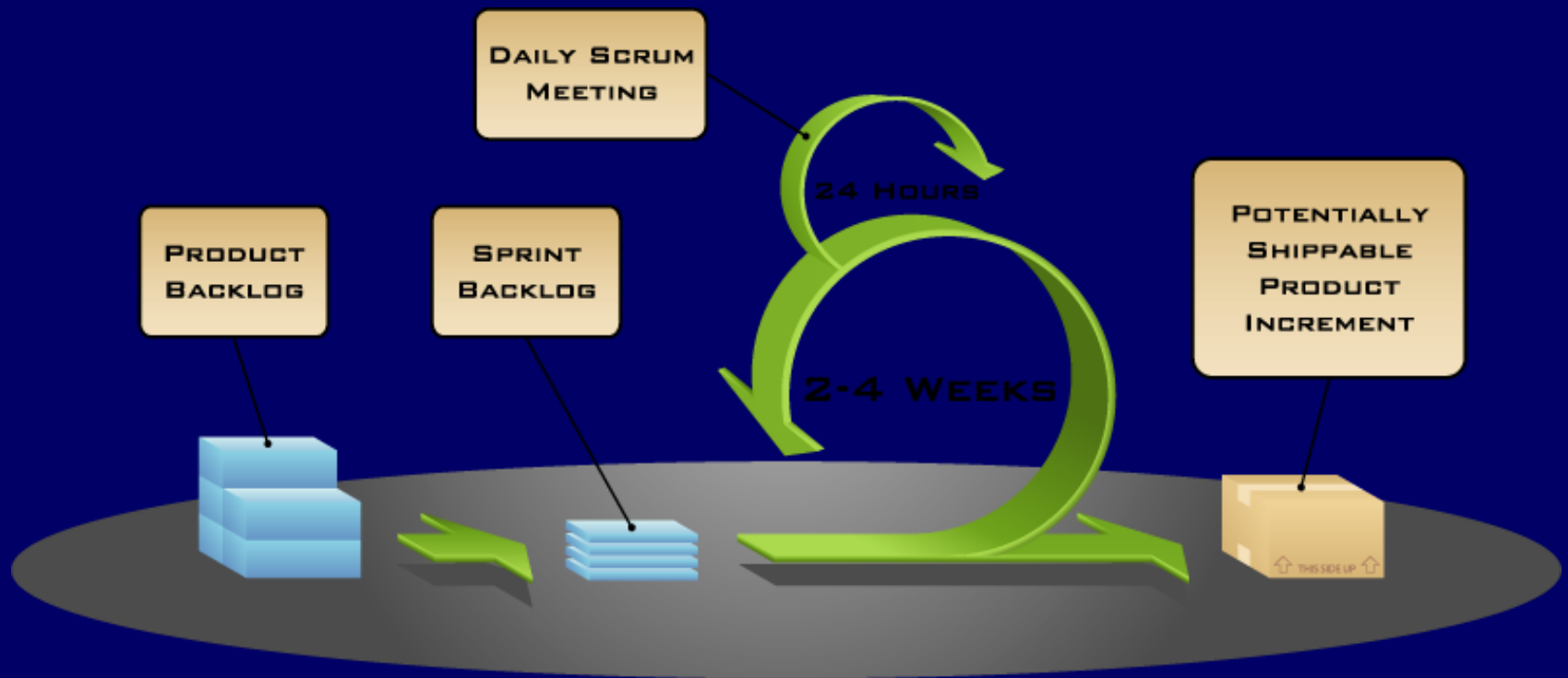
sprint

- A period of approximately 30 days in which an agreed amount of work will be performed to create a deliverable.
- ▣ SCRUM's goal is to deliver as much quality software as possible within a series (three to eight) of Sprints
 - Analogous to XP iterations
- ▣ A sprint is the basic unit of development in Scrum
- ▣ Product is designed, coded, and tested during the sprint
- ▣ **NO outside influence** can interfere with the Scrum team during the Sprint

How Scrum Works?



How Scrum Works?



How does SCRUM work?

- ▣ “SCRUM Meetings” drive the completion of the allocated activities
- ▣ Each sprint operates on a number of work items called “Backlog”
- ▣ No more items are externally added into the Backlog within a Sprint, where as derived work items can be added.

How does SCRUM work?

- ▣ During each sprint, the team creates finished portions of a product. The set of features that go into a sprint come from the product *backlog*, which is a prioritized list of requirements.
- ▣ Which backlog items go into the sprint (the sprint goals) is determined during the sprint **planning meeting**. During this meeting, the Product Owner informs the team of the items in the product backlog that he or she wants completed (the ones with the highest priority).

How does SCRUM work?

- ▣ The team then determines how much of this they can commit to complete during the next sprint, and records this in the sprint backlog
- ▣ The sprint backlog is property of the development team, i.e. during a sprint, no one is allowed to edit the sprint backlog except for the development team.

Artifacts

▣ Artifacts

- 1) Product Backlog,
- 2) Sprint Goals
- 3) Sprint Backlog,
- 4) Burndown Chart

Product Backlog

- ▣ A list of all desired work on the project
- ▣ List is prioritized by the Product Owner
 - Typically a Product Manager, Marketing, Internal Customer, etc.

Product Backlog

- ▣ Requirements for a system, expressed as a prioritized list of build up Items
- ▣ Is managed and owned by a Product Owner
- ▣ Spreadsheet (typically)
- ▣ Usually is created during the Sprint Planning Meeting
- ▣ Can be changed and re-prioritized before each Project meeting

Sample Product Backlog

	Item #	Description	Est	By
Very High				
	1	Finish database versioning	16	KH
	2	Get rid of unneeded shared Java in database	8	KH
		- Add licensing	-	-
	3	Concurrent user licensing	16	TG
	4	Demo / Eval licensing	16	TG
		Analysis Manager		
	5	File formats we support are out of date	160	TG
	6	Round-trip Analyses	250	MC
High				
		- Enforce unique names	-	-
	7	In main application	24	KH
	8	In import	24	AM
		- Admin Program	-	-
	9	Delete users	4	JM
		- Analysis Manager	-	-
	10	When items are removed from an analysis, they should show up again in the pick list in lower 1/2 of the analysis tab	8	TG
		- Query	-	-
	11	Support for wildcards when searching	16	T&A
	12	Sorting of number attributes to handle negative numbers	16	T&A
	13	Horizontal scrolling	12	T&A
		- Population Genetics	-	-
	14	Frequency Manager	400	T&M
	15	Query Tool	400	T&M
	16	Additional Editors (which ones)	240	T&M
	17	Study Variable Manager	240	T&M
	18	Haplotypes	320	T&M
	19	Add icons for v1.1 or 2.0	-	-
		- Pedigree Manager	-	-
	20	Validate Derived kindred	4	KH
Medium				
		- Explorer	-	-
	21	Launch tab synchronization (only show queries/analyses for logged in users)	8	T&A
	22	Delete settings (?)	4	T&A

Sprint Goals

- ▣ The sprint goal summarizes the desired outcome of the sprint.
- ▣ It should move the Scrum team a step closer toward the launch of a successful product.
- ▣ Sprint goals are the result of a negotiation between the product owner and the development team.
- ▣ A Sprint goal might be along the lines of:
"This Sprint we will allow users to log-in to the site, retrieve a forgotten password, and manage their own profile".

Sprint Goals

- ▣ A sprint goal is a short, one- or two-sentence, description of what the team plans to achieve during the sprint.
- ▣ . The following are example sprint goals on an eCommerce application:
 - “ - Implement basic shopping cart functionality including add, remove, and update quantities.
 - Develop the checkout process: pay for an order, pick shipping, order gift wrapping, etc.”

Sprint Goals

- ▣ Sprint goals are beneficial for several reasons:
 - They create alignment among the product owner, ScrumMaster, and team: Everyone is working toward a common goal.
 - They minimize variation by limiting the type of requirements worked on in a given sprint, for instance, by choosing items from the same theme. This facilitates close teamwork and can help increase velocity.
 - They make it easier to communicate to stakeholders what the team is working on.

Sprint Goals

- ▣ The sprint goals should not be changed during the sprint. Development is timeboxed such that the sprint must end on time; if requirements are not completed for any reason they are left out and returned to the product backlog. After a sprint is completed, the team demonstrates how to use the software.

Sprint Backlog during the Sprint

▣ Changes

- Team adds new tasks whenever they need to in order to meet the Sprint Goal
- Team can remove unnecessary tasks
- But: Sprint Backlog can only be updated by the team

▣ Estimates are updated whenever there's new information

Sprint Backlog

- ▣ A subset of Product Backlog Items, which define the work for a Sprint
- ▣ Is created ONLY by Team members
- ▣ Each Item has it's own status
- ▣ Should be updated every day

Sprint Backlog

- ▣ No more than 300 tasks in the list
- ▣ If a task requires more than 16 hours, it should be broken down
- ▣ Team can add or subtract items from the list. Product Owner is not allowed to do it

Sample Sprint Backlog

		Days Left in Sprint	15	13	10	8	
Who	Description		7/22/2002	7/24/2002	7/26/2002	7/31/2002	
Total Estimated Hours:		554	458	362	270	0	
-	User's Guide	-	-	-	-	-	
SM	Start on Study Variable chapter first draft	16	16	16	16		
SM	Import chapter first draft	40	24	6	6		
SM	Export chapter first draft	24	24	24	6		
	Misc. Small Bugs						
JM	Fix connection leak	40					
JM	Delete queries	8	8				
JM	Delete analysis	8	8				
TG	Fix tear-off messaging bug	8	8				
JM	View pedigree for kindred column in a result set	2	2	2	2		
AM	Derived kindred validation	8					
	Environment						
TG	Install CVS	16	16				
TBD	Move code into CVS	40	40	40	40		
TBD	Move to JDK 1.4	8	8	8	8		
	Database						
KH	Killing Oracle sessions	8	8	8	8		
KH	Finish 2.206 database patch	8	2				
KH	Make a 2.207 database patch	8	8	8	8		
KH	Figure out why 461 indexes are created	4					

Burn down Charts

- ▣ Are used to represent “work done”.
- ▣ Are wonderful Information Radiators
- ▣ 3 Types:
 - Sprint Burn down Chart (progress of the Sprint)
 - Release Burn down Chart (progress of release)
 - Product Burn down chart (progress of the Product)

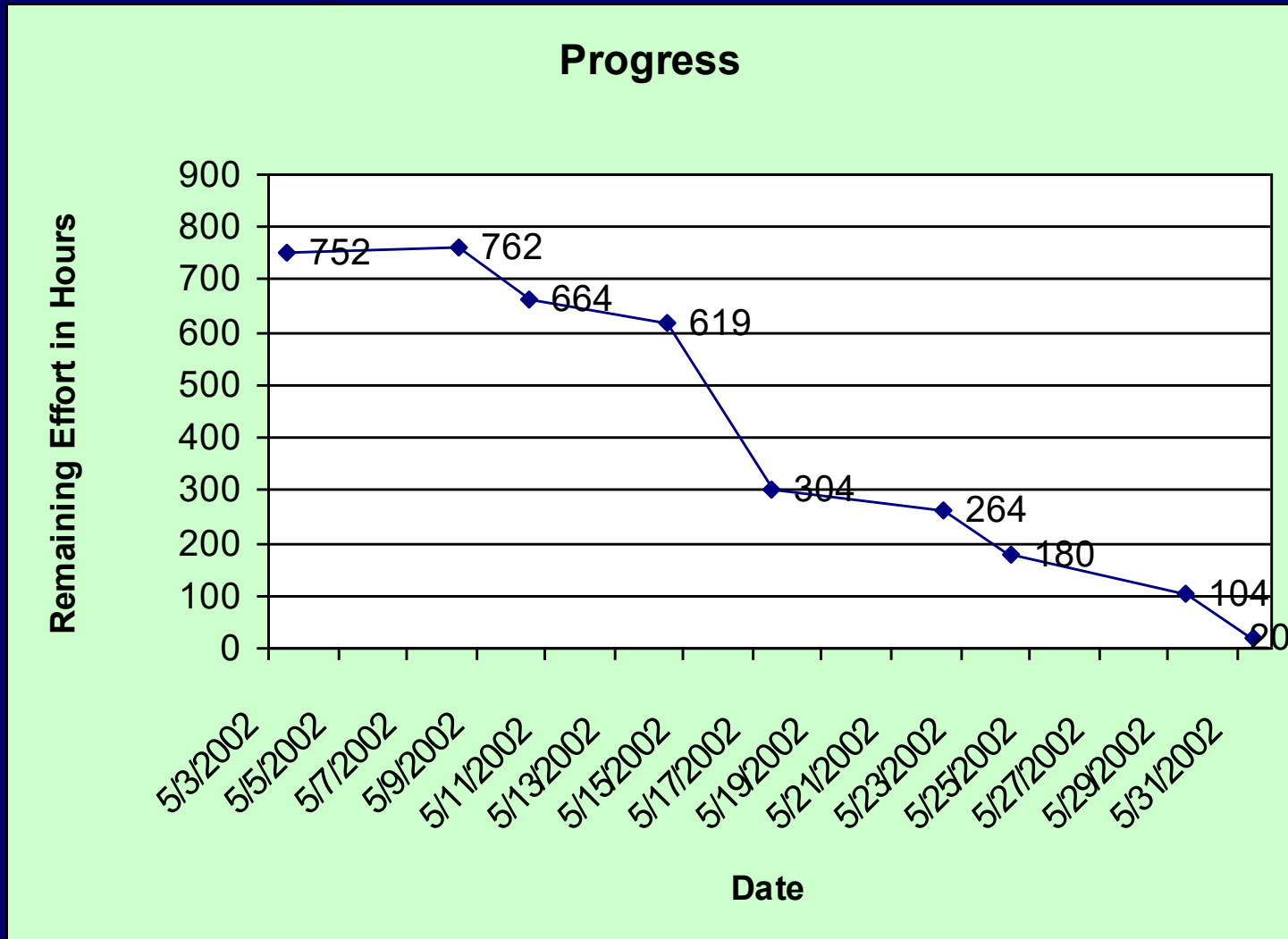
Burn down Charts

- ▣ X-Axis: time (usually in days)
- ▣ Y-Axis: remaining effort

Sprint Burn down Chart

- ▣ Depicts the total Sprint Backlog hours remaining per day
- ▣ Shows the estimated amount of time to release
- ▣ Ideally should burn down to zero to the end of the Sprint
- ▣ Actually is not a straight line
- ▣ Can bump UP

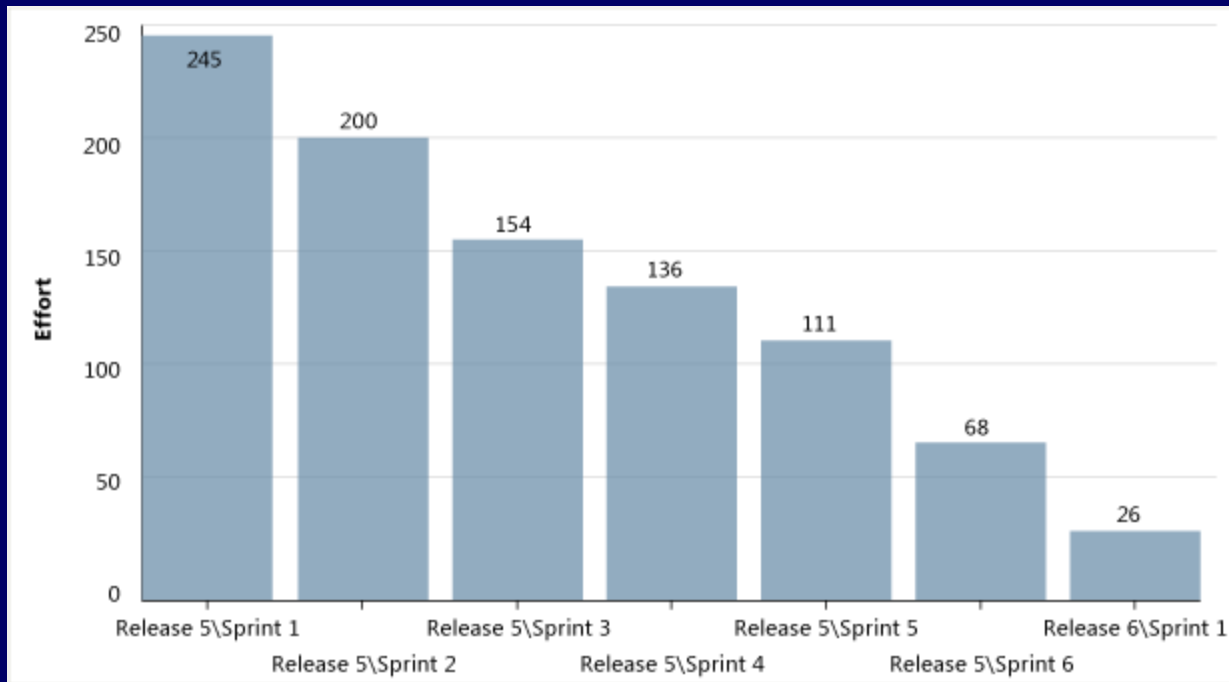
Sprint Burndown Chart



Release Burn down Chart

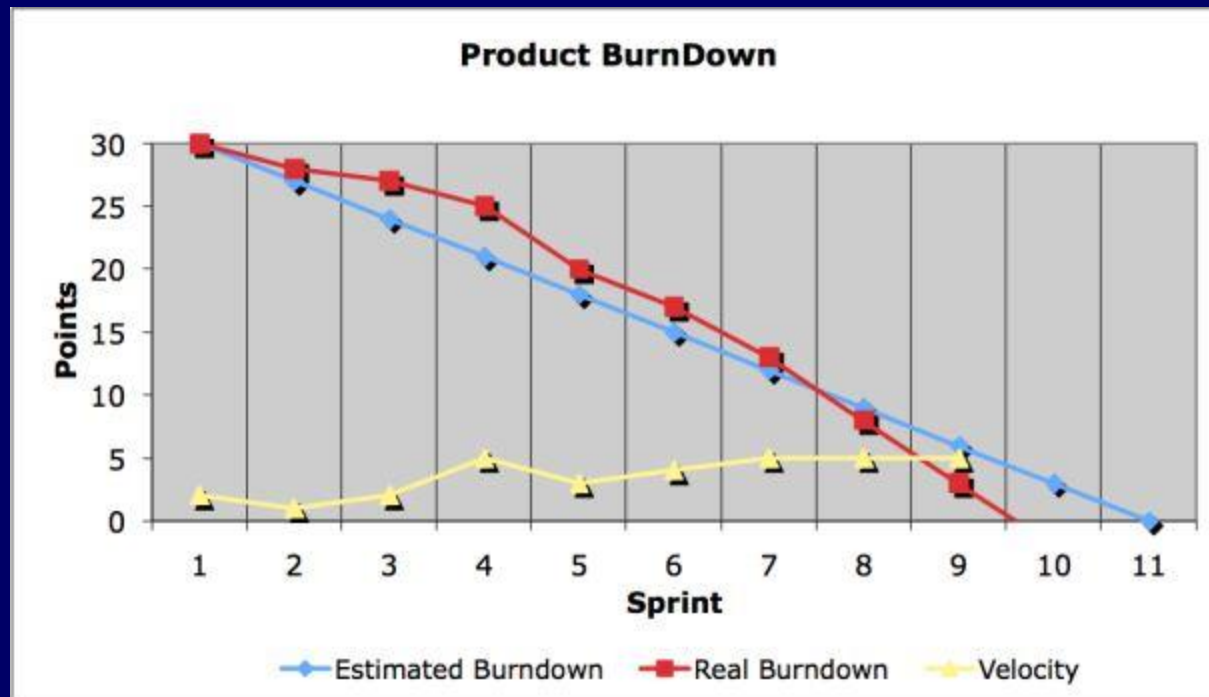
- ▣ Will the release be done on right time?
- ▣ X-axis: sprints
- ▣ Y-axis: amount of hours remaining
- ▣ The estimated work remaining can also burn up
- ▣ The release burndown chart is updated at the end of each sprint by the ScrumMaster

Release Burn down Chart

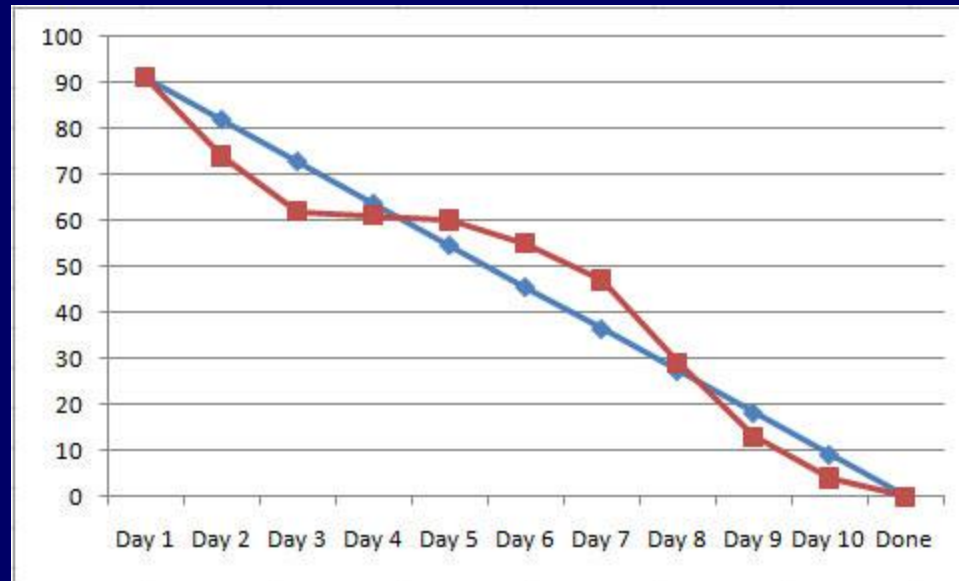


Product Burn down Chart

- ▣ Is a “big picture” view of project’s progress (all the releases)



Product Burn down Chart



The Process

- ▣ **The Process:**
- ▣ Sprint Planning,
- ▣ Sprint Review,
- ▣ Sprint presentation
- ▣ Daily Scrum Meeting

Sprint Planning Meeting

- ▣ A collaborative meeting in the beginning of each Sprint between the Product Owner, the Scrum Master and the Team
- ▣ Takes 8 hours and consists of 2 parts (“before lunch and after lunch”)

Sprint Planning

Sprint planning meeting

Sprint prioritization

- Analyze and evaluate product backlog
- Select sprint goal

Sprint goal

Sprint planning

- Decide how to achieve sprint goal (design)
- Create sprint backlog (tasks) from product backlog items (user stories / features)
- Estimate sprint backlog in hours

Sprint backlog

Team capacity

Product backlog

Business conditions

Current product

Technology

The Sprint Planning Meeting

1. Product Owner describes highest priority features to the Team.
2. Team decides what the can commit to delivering in the Sprint.

Parts of Sprint Planning Meeting

- ▣ 1st Part:
 - Creating Product Backlog
 - Determining the Sprint Goal.
 - Participants: Product Owner, Scrum Master, Scrum Team

Parts of Sprint Planning Meeting

- ▣ 2nd Part:
 - Participants: Scrum Master, Scrum Team
 - Creating Sprint Backlog

From Sprint Goal to Sprint Backlog

- ▣ Scrum team takes the Sprint Goal and decides what tasks are necessary
- ▣ Team self-organizes around how they'll meet the Sprint Goal
 - Manager doesn't assign tasks to individuals
- ▣ Managers don't make decisions for the team
- ▣ Sprint Backlog is created

Sprint Review Meeting

- ▣ Is held at the end of each Sprint
- ▣ Business functionality which was created during the Sprint is demonstrated to the Product Owner
- ▣ Informal, should not divert Team members of doing their work

Sprint Review Meeting

- ▣ Team presents what it accomplished during the sprint
- ▣ Typically takes the form of a demo of new features or underlying architecture
- ▣ Informal
 - 2-hour prep time rule
- ▣ Participants
 - Customers
 - Management
 - Product Owner
 - Other engineers

Daily Scrum

- ▣ Is a short (15 minutes long) meeting, which is held every day before the Team starts working
- ▣ Participants: Scrum Master (which is the chairman), Scrum Team only.

The Daily Scrum

- ▣ Every Team member should answer on following questions:
 1. What have you accomplished since yesterday?
 2. Are your Sprint Backlog estimates accurate?
 3. What are you working on today?
 4. Is there anything blocking you?

Daily Scrum

- ▣ Is NOT a problem solving session
- ▣ Is NOT a way to collect information about WHO is behind the schedule
- ▣ Is a meeting in which team members make commitments to each other and to the Scrum Master
- ▣ Is a good way for a Scrum Master to track the progress of the Team

XP and Scrum

Scrum and extreme programming (XP) are both Agile methodologies

- ▣ *Scrum* is a product development methodology consisting of practices and rules to be used by management, customers, and project management to maximize the productivity and value of a development effort.
- ▣ Scrum takes the responsibility for development projects out of engineering and IT and puts them directly back in the business.

XP and Scrum

- ▣ A business uses Scrum to run development projects in a businesslike manner, paying particular attention to realizing the value of the investments as soon as possible.
- ▣ Scrum doesn't have any engineering practices, covering and using those at the organization where it is implemented.

XP and Scrum

- ▣ *Extreme programming (XP) is an engineering methodology consisting of practices that ensure top-quality, focused code.*
- ▣ XP doesn't have any management practices. XP tells management where it needs them, but offers few insights into maximizing value.

XP and Scrum

- ▣ Scrum and extreme programming provide balancing practices and rules.
- ▣ They overlap at the planning game (XP) and Sprint planning (Scrum). Both encourage similar values, minimizing otherwise hard disconnects between management and developers.
- ▣ Combined, they provide a structure within which a customer can evolve a software product that best meets his or her needs, and can implement quality functionality incrementally to take advantage of business opportunities

XP and Scrum

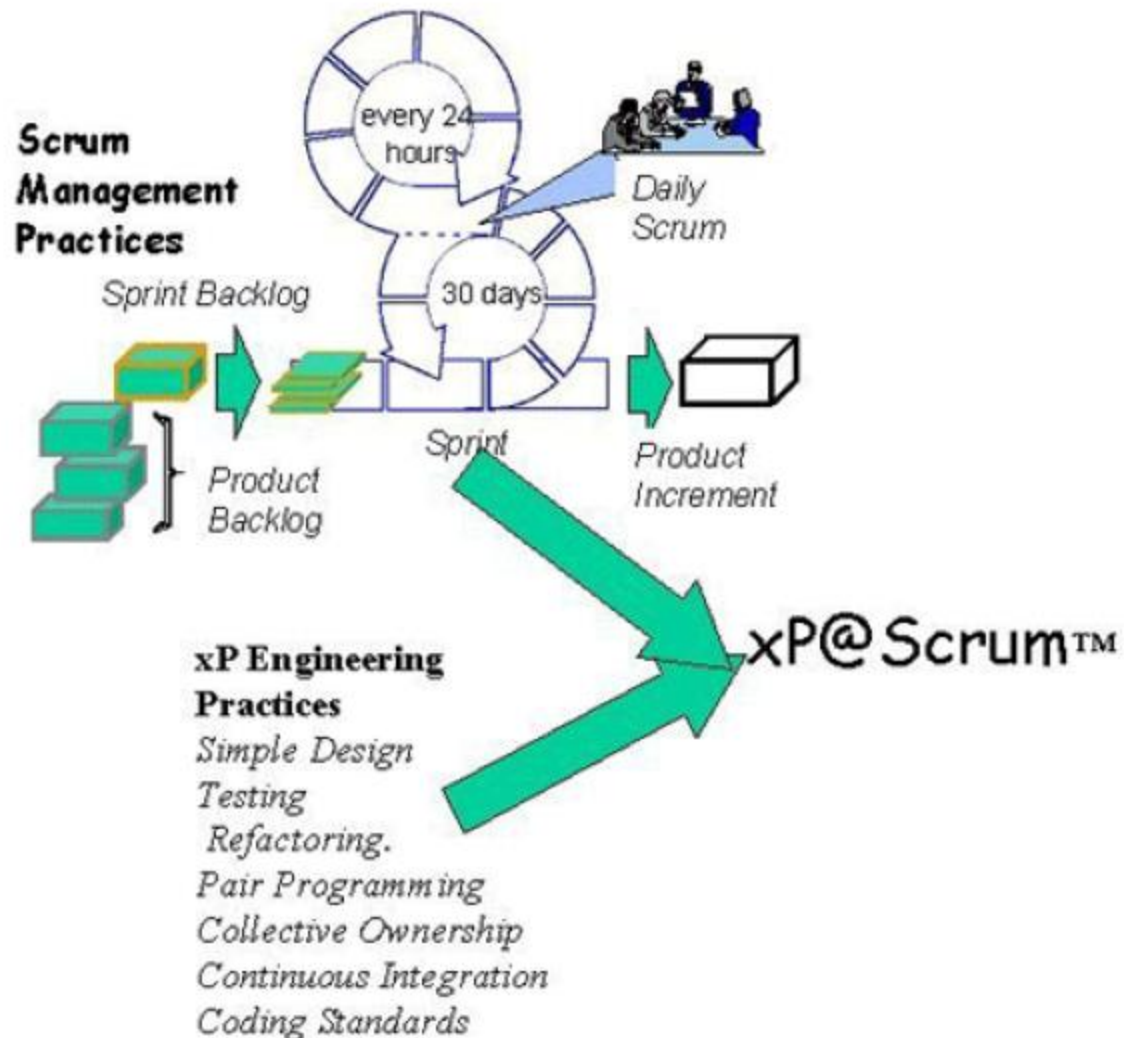


Figure 1 Scrum and extreme programming.